

Aldehyde dehydrogenase 2 (ALDH2) is essential for the oxidation of toxic aldehydes and uses NAD^+ as the oxidizing agent. Mitochondrial ALDH2 exists as a 220-kD homotetrameric protein containing a catalytic, nucleotide-binding, and coenzyme-binding domain. The most prevalent ALDH2 mutation (ALDH2*2) involves an E to K (E478K) substitution in the catalytic domain that leads to decreased enzymatic activity.

Scientists hypothesize that Asian populations have higher rates of ALDH2*2 incidence. The properties of ALDH2 variants in Caucasian and Asian populations were analyzed in the following experiments.

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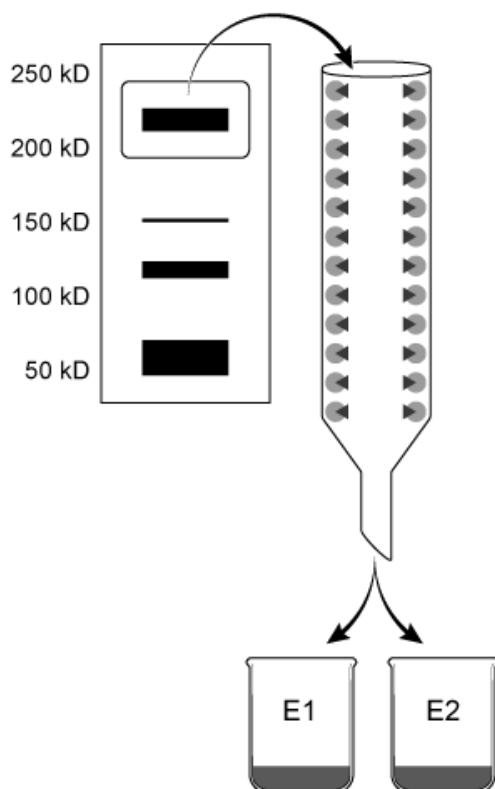


Figure 1 Purification of ALDH2 proteins

ALDH2 activity in each eluate was quantified by adding acetaldehyde and NAD^+ and then monitoring the rate of NADH production. The results are shown in Figure 2.

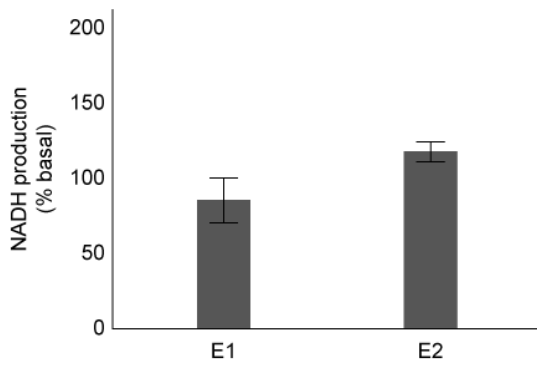


Figure 2 Enzymatic activity of E1 and E2 eluates

Experiment 2

Researchers predict that mutations at conserved amino acid sites containing S or T may prevent ALDH2 from being phosphorylated by protein kinase C epsilon (ϵ PKC). The activity of wild-type ALDH2 (WT) and ALDH2 mutants with substitutions at positions T185 and S279 in the presence or absence of ϵ PKC are summarized in Table 1.

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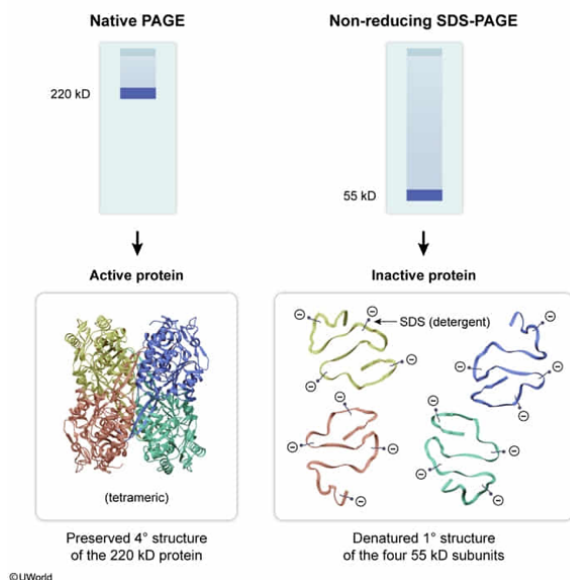
Replacing native gel electrophoresis with nonreducing sodium dodecyl sulfate (SDS) polyacrylamide gel electrophoresis (PAGE) during the purification step results in eluates with no enzymatic activity. The best explanation of this observation would be:

- ☒ A. ALDH2 must retain its quaternary structure to catalyze reactions. [59%]
- ☐ B. the SDS-PAGE breaks disulfide bonds between ALDH2 subunits. [29%]
- ☐ C. ALDH2 has the mobility of a 55-kD protein in native PAGE. [4%]
- ☐ D. the primary structure of the ALDH2 active site has been modified. [6%]

Correct

59% answered correctly

Explanation:



ALDH2 proteins were initially isolated by **native polyacrylamide gel electrophoresis** (PAGE), which is used to separate proteins by charge and molecular weight without affecting their three-dimensional structure. In this procedure, an electric field is applied to a gel and proteins migrate toward one of the electrodes based on their mass and charge. Under these conditions, proteins maintain their biological function because they retain their tertiary and quaternary structure during native PAGE purification.

In contrast, **SDS-PAGE** separates proteins only on the basis of molecular weight. To accomplish this, proteins are coated with sodium dodecyl sulfate (SDS), a detergent that confers a negative charge. Under these conditions, each protein in the gel has a similar charge and migrates toward the anode. In some cases, **reducing agents** may be added to SDS gels to break any disulfide bonds that are present. Although SDS-PAGE is superior for estimating the molecular weight of a protein, **detergents** such as SDS disrupt **noncovalent interactions** and cause proteins to unfold, or become denatured. As a result, proteins exist only in their **primary structure** (linear sequence of amino acids) and cannot perform their biological function.

Replacing native PAGE with SDS-PAGE results in the denaturation of active ALDH2 tetramers. The lack of enzymatic activity in eluates obtained from SDS-PAGE purification indicates that ALDH2 proteins lost their quaternary structure, preventing them from functioning.

(Choice B) Reducing agents can separate multisubunit proteins into individual subunits by cleaving disulfide bonds, but *nonreducing* SDS-PAGE does not contain reducing agents.

(Choice C) ALDH2 would migrate as a 220-kD tetramer (four subunits) in native PAGE.

(Choice D) SDS does not alter the primary structure (amino acid sequence) of proteins.

Educational objective:

Native polyacrylamide gel electrophoresis (PAGE) separates proteins by charge and mass without denaturing proteins whereas SDS-PAGE separates proteins only on the

basis of mass. Sodium dodecyl sulfate (SDS) is a detergent that coats proteins with a negative charge and denatures proteins by disrupting noncovalent interactions. Proteins separated by SDS-PAGE are unfolded into their primary structure, and are unable to perform their biological functions.

Time spent:0 sec

Subject:
Biochemistry

QID:400944

System:
Amino Acids and Proteins

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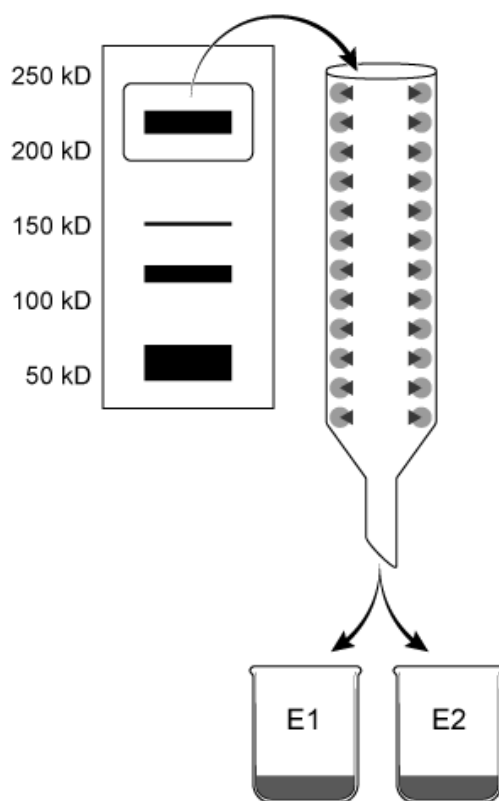


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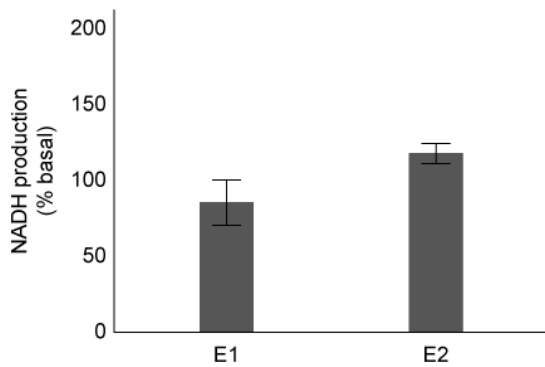


Figure 2 Enzymatic activity of E1 and E2 eluates

Experiment 2

Researchers predict that mutations at conserved amino acid sites containing S or T may prevent ALDH2 from being phosphorylated by protein kinase C epsilon (ϵ PKC). The activity of wild-type ALDH2 (WT) and ALDH2 mutants with substitutions at positions T185 and S279 in the presence or absence of ϵ PKC are summarized in Table 1.

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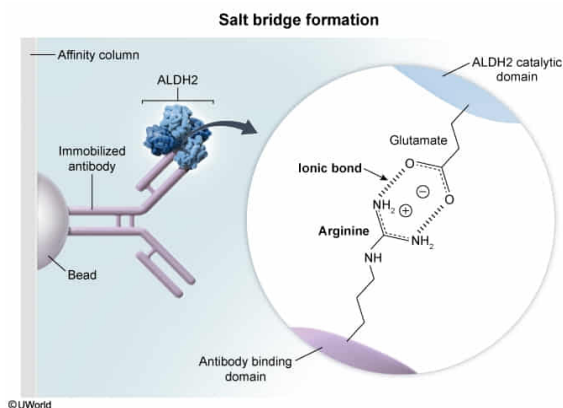
The active site of wild-type ALDH2 most likely interacts with which of the following amino acid residues on the antibodies shown in Figure 1?

- ☐ A. Aspartate [22%]
- ☐ B. Phenylalanine [3%]
- ☐ C. Isoleucine [2%]
- ✓ ☒ D. Arginine [71%]

Correct

71% answered correctly

Explanation:



Affinity chromatography is a **separation technique** that isolates proteins based on **highly specific noncovalent binding interactions**. Affinity columns can be created by attaching antibodies (proteins with high affinity for their targets) to beads known as the stationary phase, that serve as the solid support. When a mixture of proteins passes through the column, only the target protein is bound. Contaminant proteins and other molecules with low affinity for the antibody remain unbound and are washed out of the column.

The affinity column shown in Figure 1 was specifically designed to separate wild-type ALDH2 from ALDH2*2 by selectively binding the catalytic domain of wild-type ALDH2. The first buffer wash removed the unbound ALDH2*2 and collected it in E1 whereas the second wash dislodged wild-type ALDH2 from antibodies.

The catalytic domain of wild-type ALDH2 contains a negatively charged **glutamate** residue (E), which could interact with positively charged amino acids on the antibodies, such as **arginine**. Therefore, an **arginine** residue in the antibodies **most likely facilitates binding** to the active site of wild-type ALDH2. ALDH2*2, on the other hand, has positively charged **lysine** (K) in place of glutamate and will be repelled by positively charged residues on the antibody.

(Choice A) The negatively charged carboxyl group in the R group of **aspartate** would repel glutamate in the catalytic domain of wild-type ALDH2.

(Choices B and C) Phenylalanine and isoleucine are **hydrophobic amino acids** that do not participate in electrostatic interactions such as ionic bonds.

Educational objective:

Affinity chromatography separates proteins based on their specific noncovalent-binding interactions with antibodies or other ligands attached to a column. Negatively charged amino acids such as aspartate and glutamate interact with positively charged amino acids such as arginine, histidine, and lysine to form ionic bonds.

Time spent: 0 sec
Subject:
 Biochemistry

QID: 400945
System:
 Amino Acids and Proteins

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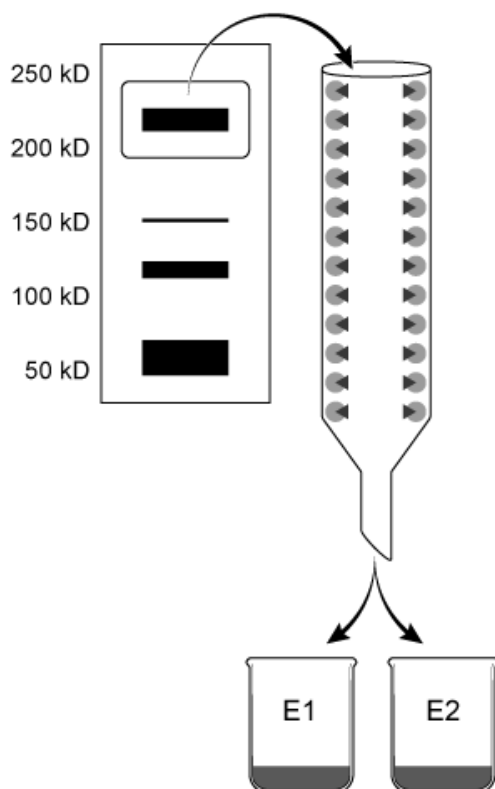


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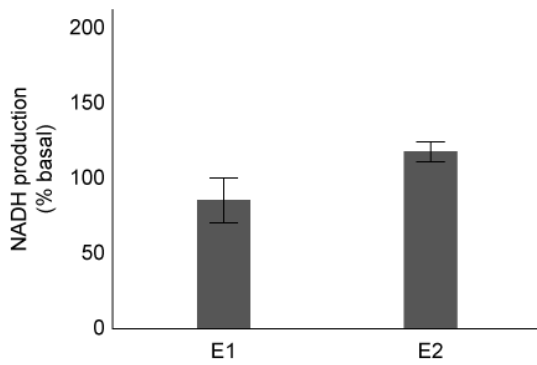


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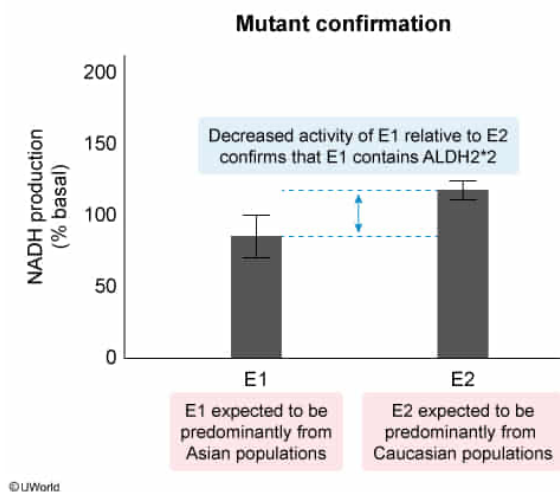
Scientists could provide support for the hypothesis that Asian populations have higher rates of ALDH2*2 mutations if they discovered which of the following?

- ☐ A. The proteins in E2 were exclusively from Asian samples. [15%]
- ✓ ☒ B. None of the proteins in E1 were from Caucasian samples. [63%]
- ☐ C. Only WT-ALDH2 proteins were eluted from the affinity column. [5%]
- ☐ D. ALDH2*2 proteins were eluted first from the affinity column. [15%]

Correct

63% answered correctly

Explanation:



Scientists hypothesized that Asian populations would have higher rates of the ALDH2*2 mutation than Caucasian populations. ALDH2*2 mutant proteins contain an E478K mutation that decreases enzymatic activity. To compare the enzymatic activity of wild-type (WT) ALDH2 and the ALDH2*2 mutant, the enzymatic assay shown in Figure 2 was performed using the proteins isolated in E1 and E2. The enzymatic activity (NADH production) of the eluate that contains ALDH2*2 proteins is expected to be lower than that of wild-type protein. According to Figure 2, the proteins in E2 were more active than those in E1, which confirms that E2 contains wild-type ALDH2 and E1 contains ALDH2*2.

Scientists could **support their hypothesis** if they discovered that ALDH2*2 mutants found in E1 predominantly belonged to Asian samples or that WT-ALDH2 proteins in E2 were obtained primarily from Caucasian samples. Therefore, if no mutant proteins from E1 came from Caucasian samples, this would be **consistent with the hypothesis** that Asian populations have a higher incidence of ALDH2*2 mutations than Caucasian populations.

(Choice A) E2 contains WT-ALDH2 proteins, which are expected to belong primarily to Caucasian samples.

(Choice C) WT-ALDH2 proteins and mutant ALDH2*2 proteins were removed from the column in separate washes but were both eluted.

(Choice D) ALDH2*2 was eluted first from the column into E1, but this does not indicate that it belonged to Asian samples.

Educational objective:

The hypothesis of an experiment must be tested to be accepted or rejected. A hypothesis is accepted when the experimental results confirm the prediction being made.

Time spent: 0 sec
 Subject:
 Biochemistry

QID: 400946
 System:
 Amino Acids and Proteins

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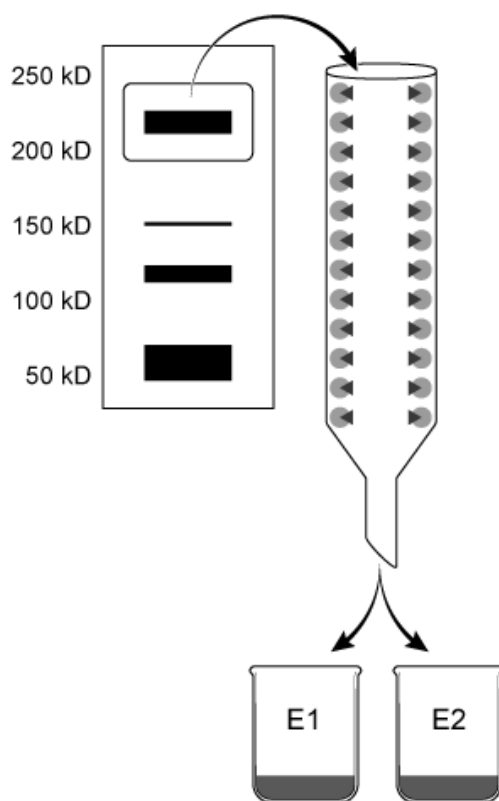


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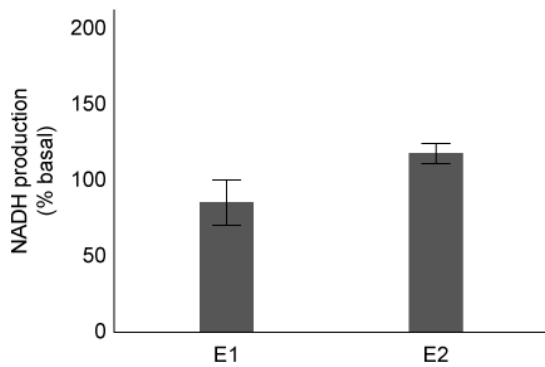


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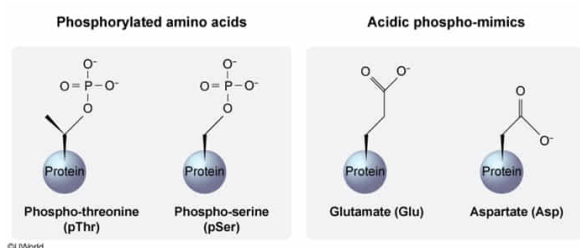
Based on Table 1, which pair of amino acid substitutions in ALDH2 would most likely result in the highest NADH production in the absence of ϵ PKC?

- ☐ A. T185N and S279Q [33%]
- ☐ B. T185D and S279N [6%]
- ☒ C. T185E and S279D [53%]
- ☐ D. T185Q and S279E [6%]

Correct

52% answered correctly

Explanation:



Phosphorylation, a type of protein post-translational modification, adds a phosphate group to **serine**, **threonine**, or **tyrosine** residues. These residues are neutrally charged at physiological pH, but the addition of a phosphate group confers **negative charge**. This negative charge can alter the activity of a protein, increasing or decreasing its activity. Substituting an amino acid with a negative charge (**glutamate or aspartate**) into a protein can mimic the **effects of phosphorylation**.

Table 1 shows that in the absence of ϵ PKC, ALDH2 has half the activity as it does in the presence of ϵ PKC. In addition, mutating certain threonine or serine residues (T185 or S279) to alanine (which cannot be phosphorylated) reduces the activity even when ϵ PKC is present. This indicates that ϵ PKC upregulates ALDH2 activity by phosphorylating T185 and S279.

If glutamate (E) or aspartate (D) were substituted into positions 185 and 279, the negative charges they carry would likely mimic the effects of negatively charged phosphate. Therefore, a **T185E and S279D** mutation would most likely increase ALDH2 activity (via the same effects as phosphorylation) when ϵ PKC is absent, resulting in the **highest NADH production**.

(Choices A, B, and D) **Asparagine** (N) and **glutamine** (Q) are similar in structure to aspartate and glutamate, respectively, but are not negatively charged. They would not mimic the effect of phosphorylated serine or threonine, and therefore would likely not result in high NADH production when ϵ PKC is absent.

Educational objective:

Phosphorylation of serine, threonine, and tyrosine residues adds negative charge to a protein, and often alters its activity. The effect of phosphorylation can be mimicked by the negative charges of aspartate or glutamate substituted at positions where phosphorylation normally occurs.

Time spent: 0 sec
Subject:
Biochemistry

QID: 400947
System:
Amino Acids and Proteins

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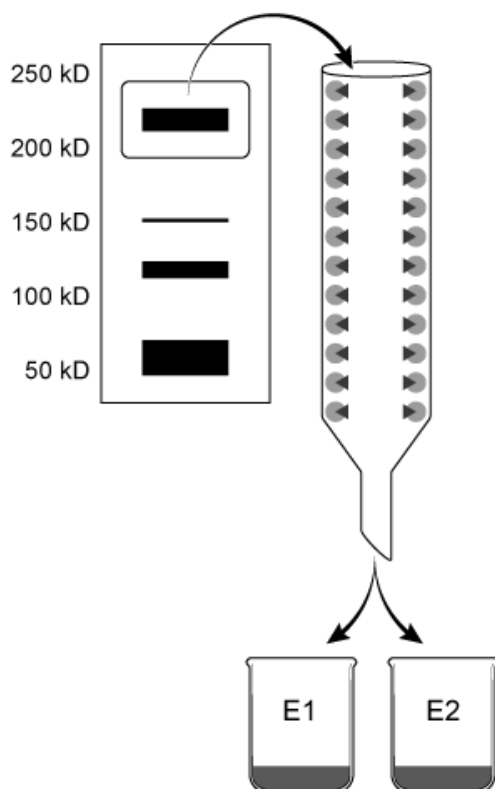


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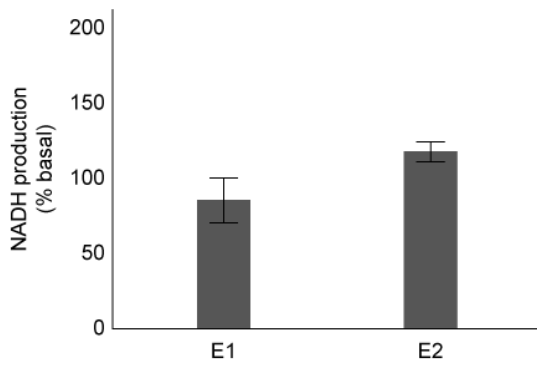


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Increasing the concentration of sodium chloride (NaCl) in the E2 eluate could affect the rate of NADH production in what way?

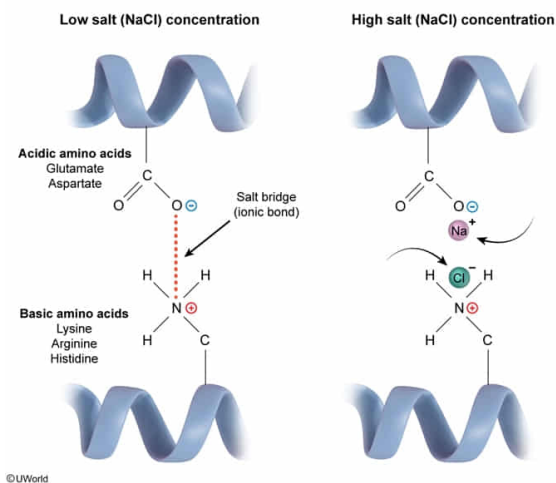
- ☐ A. Increased NADH production by enhancing hydrogen bonding [5%]
- ☐ B. Decreased NADH production by reducing hydrophobicity [8%]
- ☐ C. Increased NADH production by increasing basicity [10%]
- ☒ D. Decreased NADH production by disrupting ionic bonding [76%]

Correct

75% answered correctly

Explanation:

Electrostatic interactions in proteins



Electrostatic interactions such as ionic bonds are noncovalent forces that help maintain the tertiary and quaternary structure of a protein. They occur mainly between the acidic and basic side chains of amino acids on the polypeptide chain, and are commonly referred to as **salt bridges**. Neutral salts such as NaCl, which dissociate in solution, can disrupt salt bridges by forming ionic bonds with amino acids. Positively charged sodium ions (Na⁺) associate with the negatively charged side chains of **acidic amino acids**. Chloride ions (Cl⁻), which are negatively charged, associate with the positively charged side chains of **basic amino acids**.

High concentrations of salt can cause a protein to unfold through a process known as **denaturation**. The accompanying loss of three-dimensional structure also results in a **loss or reduction of function**. Under controlled physiological conditions, ALDH2 produces NADH during oxidation-reduction reactions. However, ALDH2 proteins exposed to increasing concentrations of salt in the eluate could become denatured, resulting in a decrease in the rate of NADH production.

(Choice A) NaCl can interfere with hydrogen bonding in the secondary structure of a protein but cannot enhance it. As with ionic bonds, interrupting **hydrogen bonds** within ALDH2 could also lead to denaturation and a reduction in NADH production.

(Choice B) NaCl does not alter the hydrophobicity of a protein. Hydrophobicity is determined by the number and location of hydrophobic amino acids in the protein's primary structure.

(Choice C) NaCl would not increase the basicity because it is a neutral salt that does not contribute to a net change in hydrogen or hydroxide ion concentration.

Educational objective:

The acidic and basic side chains of amino acids form ionic bonds referred to as salt bridges, which contribute to tertiary and quaternary structure. High salt concentrations can disrupt salt bridges and cause proteins to become denatured (unfold). The loss of three-dimensional structure in denatured proteins results in a loss or reduction of function.

Time spent:0 sec

Subject:
Biochemistry

QID:400948

System:
Amino Acids and Proteins

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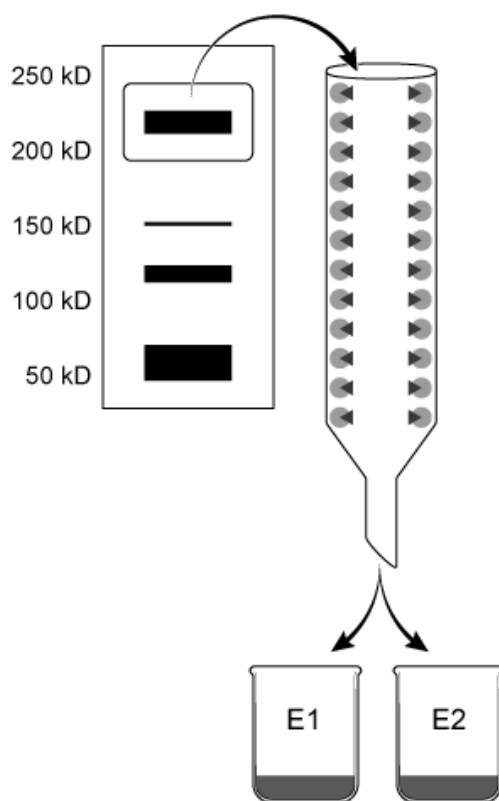


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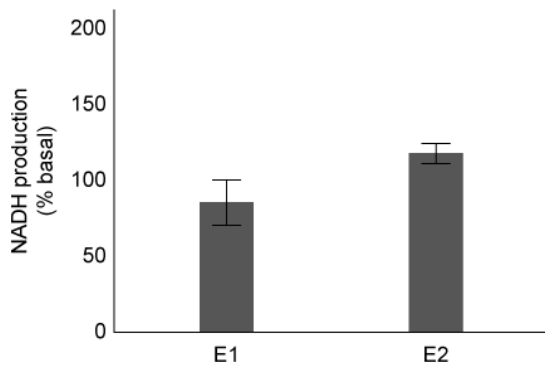


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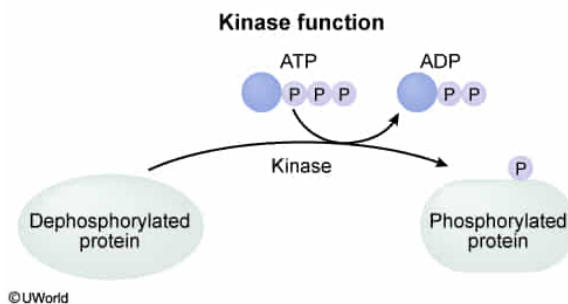
The method used to phosphorylate ALDH2 in Experiment 2 must include ϵ PKC and which other component?

- ☐ A. Inorganic phosphate [20%]
- ☒ B. ATP [74%]
- ☐ C. Phosphoprotein [3%]
- ☐ D. Glucose-6-phosphate [1%]

Correct

74% answered correctly

Explanation:



Kinases are enzymes that catalyze the **transfer of phosphate groups** from nucleotide triphosphates (ie, **ATP**, **GTP**) to other molecules or from phosphorylated molecules to nucleotide diphosphates (ie, ADP, GDP). According to the passage, ϵ PKC is a kinase that phosphorylates proteins such as ALDH2 at certain serine and threonine residues.

Attaching phosphate to a molecule is usually a thermodynamically unfavorable process and must be coupled to a thermodynamically favorable process to provide the necessary energy. Hydrolysis of ATP or GTP is thermodynamically favorable. For this reason, kinases use ATP or GTP as a source of phosphate in the reactions they catalyze. Therefore, in addition to ϵ PKC, **ATP must also be present** for phosphorylation of ALDH2 to occur.

(Choice A) Kinases do not transfer inorganic phosphate to molecules. The phosphate must come from a high-energy nucleotide triphosphate such as ATP or GTP.

(Choice C) ALDH2 *becomes* a phosphoprotein once phosphorylation occurs. Prior to phosphorylation, no phosphoproteins are present in this scenario.

(Choice D) Glucose-6-phosphate is produced by hexokinase during glycolysis. It is not a requirement for ϵ PKC activity.

Educational objective:

Kinases are proteins that transfer phosphate groups from ATP or GTP to other molecules or from phosphorylated molecules to ADP or GDP. For a kinase-catalyzed reaction to proceed, ATP or GTP must be present in solution.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400949

System:
Amino Acids and Proteins